T4 — Genetic Information, Variation & Relationships

4.1 DNA, Genes, and Chromosomes

Gene: A sequence of DNA nucleotides that codes for a functional polypeptide (or functional RNA molecule such as tRNA or rRNA). Genes occupy specific positions called **loci** (singular: locus) on chromosomes.

Genome: The entire DNA sequence of an organism, including both coding and non-coding regions. In eukaryotes, much of the genome (over 98% in humans) is non-coding. The Human Genome Project (completed 2003) sequenced all ~3.2 billion base pairs.

Proteome: The full range of proteins produced by a cell, tissue, or organism. The proteome is larger than the number of genes because of alternative splicing and post-translational modifications. The proteome varies between cell types even though the genome is the same.

Non-coding DNA includes:

- **Introns:** Non-coding sequences within genes, transcribed into pre-mRNA but removed during mRNA processing (splicing). Introns allow alternative splicing, producing different proteins from the same gene.
- **Regulatory sequences:** Promoters (where RNA polymerase binds), enhancers and silencers that control when, where and how much a gene is expressed.
- **Repetitive sequences / satellite DNA:** Short tandem repeats (STRs) and variable number tandem repeats (VNTRs) used in DNA fingerprinting. Their high variability between individuals makes them ideal for forensic identification.

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- **Telomeres:** Repetitive sequences (TTAGGG in humans) at chromosome ends that protect against degradation during replication. They shorten with each division; when critically short, the cell enters senescence. The enzyme telomerase can extend telomeres (active in stem cells and cancer cells).
- **Genes for non-coding RNA:** tRNA and rRNA genes do not code for polypeptides but are essential for translation.

Chromosome Structure

A chromosome is one very long DNA molecule associated with histone proteins. The packaging hierarchy is:

1. DNA double helix wraps around histone octamers (groups of 8 histone proteins), forming **nucleosomes** (the "beads on a string" structure).
2. Nucleosomes coil into a 30 nm fibre (solenoid).
3. Further looping and coiling produces **chromatin**, which exists as euchromatin (loosely packed, transcriptionally active) or heterochromatin (tightly packed, transcriptionally silent).
4. During cell division, chromatin condenses maximally into visible **chromosomes** — this ensures accurate segregation.

Homologous chromosomes are a pair of chromosomes (one maternal, one paternal) that carry the same genes at the same loci but may have different alleles. In diploid organisms (2n), chromosomes exist in homologous pairs.

> **Allele:** An alternative form of a gene. Different alleles arise by mutation and differ in their base sequence, potentially coding for different versions of the same polypeptide.

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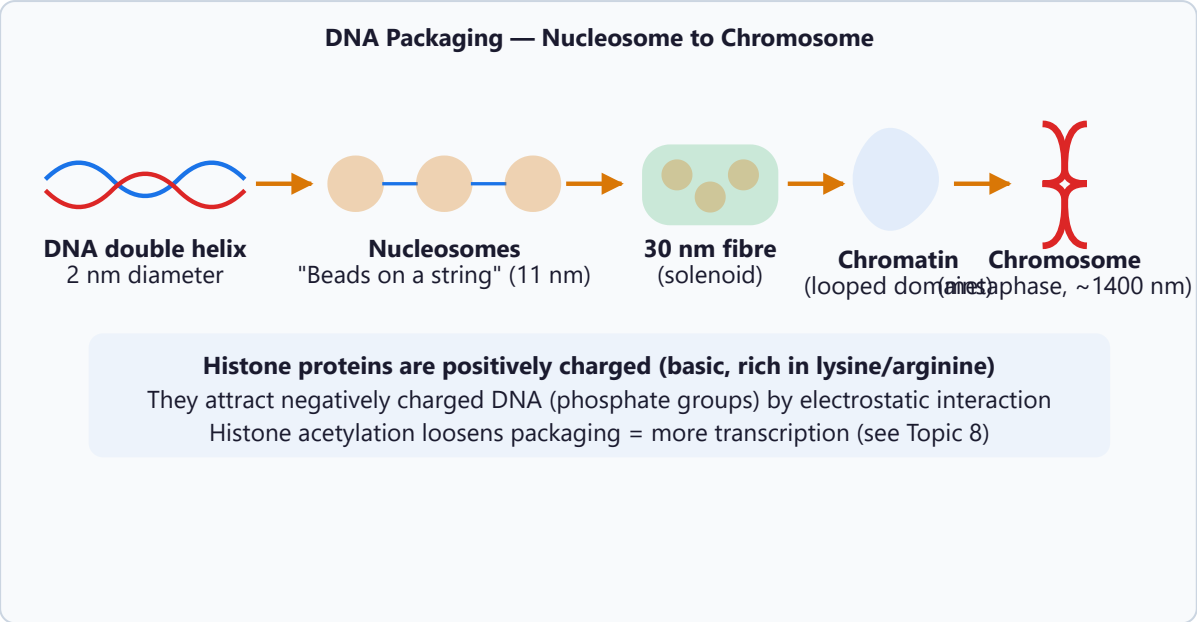


Exam Tip: Learn the difference between genome and proteome. A key AQA point: all body cells contain the same genome, but different cells express different genes (differential gene expression), so their proteomes differ. A liver cell and a neurone have the same DNA but very different proteins.

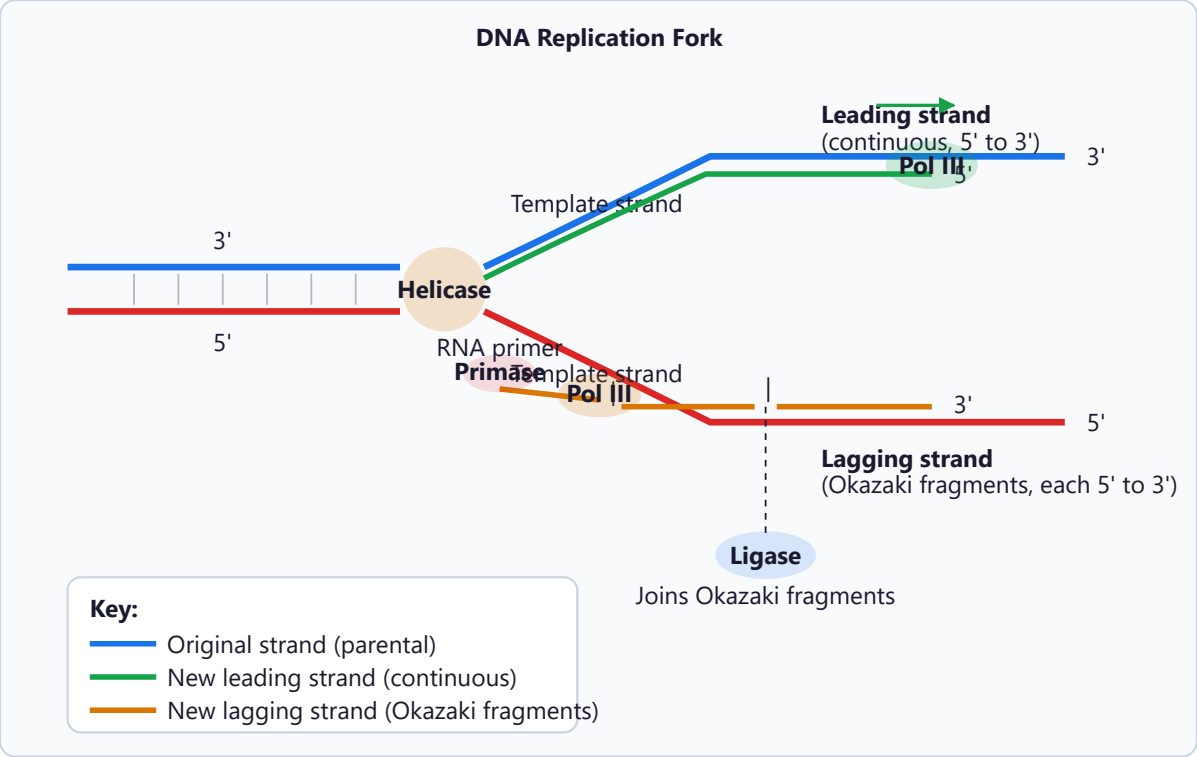
4.2 DNA Replication

Semi-conservative replication: Each new DNA molecule consists of one original (parental) strand and one newly synthesised strand. This was demonstrated by the **Meselson-Stahl experiment** (1958) using heavy nitrogen (^{15}N) and density gradient centrifugation.

Steps of DNA Replication

1. **Helicase** unwinds the double helix and breaks hydrogen bonds between complementary base pairs, separating the two strands. This creates a **replication fork**.
2. Each exposed strand acts as a template. Free DNA nucleotides (deoxyribonucleoside triphosphates) align with their complementary bases (A-T, C-G).
3. **DNA polymerase III** catalyses the formation of phosphodiester bonds between adjacent nucleotides on the new strand. It can only add nucleotides in the **5' to 3' direction** and requires a short RNA primer to begin (laid down by primase).
4. The **leading strand** is synthesised continuously in the same direction as the replication fork moves.

5. The **lagging strand** is synthesised discontinuously in short **Okazaki fragments** (each ~1000-2000 nucleotides in prokaryotes, ~100-200 in eukaryotes), because polymerase can only work 5' to 3' — it must repeatedly start new fragments as the fork opens.
6. **DNA ligase** joins Okazaki fragments together by forming phosphodiester bonds between them, creating a continuous strand.
7. **DNA polymerase I** replaces RNA primers with DNA nucleotides.



The Meselson-Stahl Experiment (1958)

This experiment provided evidence for semi-conservative replication:

1. E. coli bacteria were grown for many generations in a medium containing heavy nitrogen (^{15}N). All their DNA became "heavy" (both strands labelled).
2. Bacteria were transferred to a medium containing only light nitrogen (^{14}N) and allowed to divide.
3. After **one generation**: all DNA was at an intermediate density (one heavy strand + one light strand in each molecule) — this ruled out conservative replication.
4. After **two generations**: DNA showed equal amounts of intermediate and light bands — exactly as predicted by semi-conservative replication.
5. DNA density was measured by **caesium chloride density gradient centrifugation**.

Exam Tip: The Meselson-Stahl experiment is a classic AQA question. Be able to explain why after one generation all DNA was intermediate density (ruling out conservative replication), and after two generations there were equal bands of intermediate and light DNA. Draw a diagram showing the strand composition at each stage. Conservative replication would have produced one heavy and one light band after generation 1 — not intermediate.

4.3 Protein Synthesis

The Genetic Code

The genetic code has the following essential features:

- **Triplet:** Three bases (a codon) code for one amino acid. There are 4^3 = 64 possible codons.
- **Degenerate (redundant):** Most amino acids are coded for by more than one codon (64 codons for only 20 amino acids). This provides some protection against mutations — a change in the third base often still codes for the same amino acid (wobble hypothesis).
- **Non-overlapping:** Each base is read only once as part of a single codon.
- **Universal:** The same codons code for the same amino acids in almost all organisms — strong evidence for a common evolutionary origin of life.
- **Start codon:** AUG codes for methionine and initiates translation.
- **Stop codons:** UAA, UAG, UGA do not code for any amino acid — they signal the end of translation.

Transcription (in the nucleus)

1. **Initiation:** RNA polymerase binds to the **promoter region** upstream of the gene. Transcription factors are required to help RNA polymerase bind in eukaryotes.
2. **Elongation:** RNA polymerase moves along the **template (antisense) strand** in the 3' to 5' direction. It reads the template and assembles a complementary pre-mRNA strand in the 5' to 3' direction, using free RNA nucleotides (with uracil instead of thymine).
3. The mRNA produced is complementary to the template strand and has the same base sequence as the coding (sense) strand, except U replaces T.
4. **Termination:** RNA polymerase reaches a **terminator sequence** and detaches; the pre-mRNA is released.

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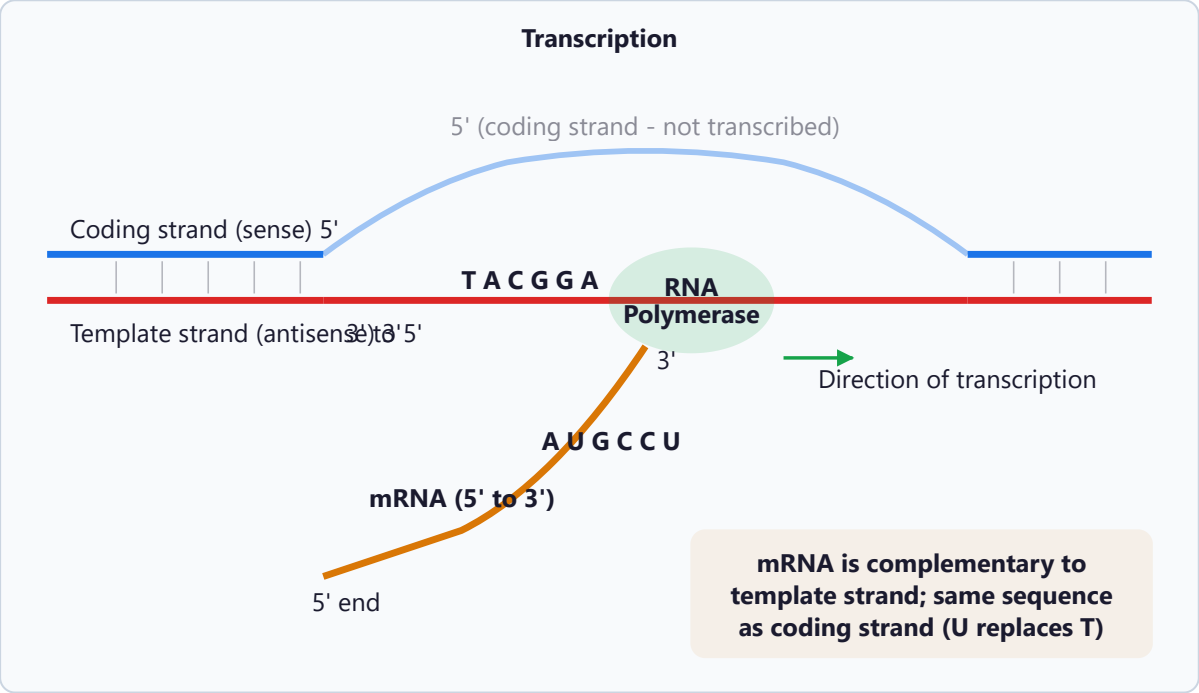
5. **Post-transcriptional modification (in eukaryotes only):**
   - Introns are spliced out by spliceosomes
   - A 5' methylguanosine cap is added (protects from degradation, aids ribosome binding)
   - A 3' poly-A tail is added (protects from enzymatic degradation, aids nuclear export)
   - **Alternative splicing:** Different combinations of exons can be joined, allowing one gene to code for multiple different proteins (this is why the human proteome is larger than the ~20,000 genes)
6. The mature mRNA exits through nuclear pores to ribosomes in the cytoplasm.



Translation (at ribosomes in the cytoplasm)

1. **Initiation:** mRNA binds to the small ribosomal subunit. The start codon (AUG) is positioned at the ribosome's **P site**. A tRNA carrying methionine (anticodon UAC) binds to AUG by complementary base pairing. The large subunit then joins.
2. **Elongation:** A second aminoacyl-tRNA enters the **A site**, with its anticodon complementary to the next mRNA codon.
3. A **peptide bond** forms between the amino acids in the P and A sites. This is catalysed by peptidyl transferase (a ribozyme — the rRNA itself acts as the enzyme).
4. The ribosome translocates one codon along the mRNA (5' to 3' direction). The tRNA in the P site moves to the **E site** and is released. The tRNA in the A site moves to the P site.

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5. A new aminoacyl-tRNA enters the A site. The cycle repeats, elongating the polypeptide chain.
6. **Termination:** When a **stop codon** (UAA, UAG, or UGA) enters the A site, no tRNA binds. Instead, a **release factor** protein binds, causing the polypeptide chain to be released. The ribosome dissociates into its subunits.

Post-translational modification: The polypeptide folds into its tertiary structure (aided by chaperone proteins). It may be modified by glycosylation, phosphorylation, or cleavage (e.g. proinsulin is cleaved to form active insulin). Many polypeptides are processed in the rough ER and Golgi apparatus.

Exam Tip: Remember that the template strand of DNA is the antisense (3' to 5') strand. The mRNA produced is complementary to this and has the same base sequence as the sense (coding) strand (with U replacing T). When given a DNA coding strand sequence, simply replace T with U to get the mRNA sequence. The anticodon on tRNA is complementary to the mRNA codon (and therefore has the same bases as the template DNA strand, with U replacing T).

tRNA Structure and Function

Transfer RNA (tRNA) is a small, single-stranded RNA molecule that folds into a characteristic **cloverleaf shape** (held by intramolecular base pairing):

- An **anticodon** (3 bases) at one end — complementary to a specific mRNA codon
- An **amino acid attachment site** (CCA sequence) at the 3' end — a specific amino acid is covalently attached by aminoacyl-tRNA synthetase (requires ATP)
- Each tRNA carries only one specific amino acid — there are at least 20 different tRNA molecules

4.4 Mutations

Gene mutation: A change in the nucleotide base sequence of DNA. This may alter the amino acid sequence of the polypeptide and therefore its tertiary structure and function. Mutations are the ultimate source of genetic variation.

Types of Point Mutation

Type	Description	Effect on Protein

Substitution	One base replaced by another	**Silent:** same amino acid (due to degenerate code, often at 3rd base position) **Missense:** different amino acid incorporated **Nonsense:** creates a premature stop codon, producing a truncated, usually non-functional protein
Insertion	One or more extra bases added	**Frameshift** — all codons downstream of the insertion are altered; usually produces a completely non-functional protein
Deletion	One or more bases removed	**Frameshift** — all codons downstream are altered; usually non-functional protein

Note: Insertions or deletions of 3 bases (or multiples of 3) add or remove whole amino acids without causing a frameshift — the rest of the protein may be unaffected.

Sickle Cell Anaemia — A Case Study

Sickle cell anaemia results from a single base substitution (point mutation) in the beta-globin gene on chromosome 11:

- DNA codon changes from **GAG to GTG** (in the template strand, CTC to CAC)
- mRNA codon changes from **GAG to GUG**
- This causes **glutamic acid** (hydrophilic, charged) to be replaced by **valine** (hydrophobic) at position 6 of the beta-globin polypeptide
- The hydrophobic valine residue causes haemoglobin S molecules to polymerise into rigid fibres under low O_2 conditions, distorting red blood cells into a sickle shape
- Sickled cells block capillaries, are fragile (haemolysis), and carry less oxygen — causing pain crises, anaemia, and organ damage

Chromosome Mutations

These involve changes in chromosome structure or number:

- **Non-disjunction:** Failure of chromosomes to separate properly during meiosis I or II, producing gametes with too many or too few chromosomes. Example: Down syndrome (trisomy 21) results from an extra copy of chromosome 21.

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- **Polyploidy:** Having more than two complete sets of chromosomes (e.g. 3n, 4n). Common in plants; can lead to speciation (sympatric speciation). Important in agriculture — many crop species are polyploid (e.g. bread wheat is hexaploid, 6n).
- **Deletions, duplications, inversions, translocations** of chromosome segments can also occur.

4.5 Meiosis and Genetic Diversity

Meiosis: A type of cell division that produces **four genetically different haploid** (n) daughter cells from one diploid (2n) parent cell. It involves two successive divisions: **meiosis I** (homologous chromosomes separate — the reduction division) and **meiosis II** (sister chromatids separate — similar to mitosis).

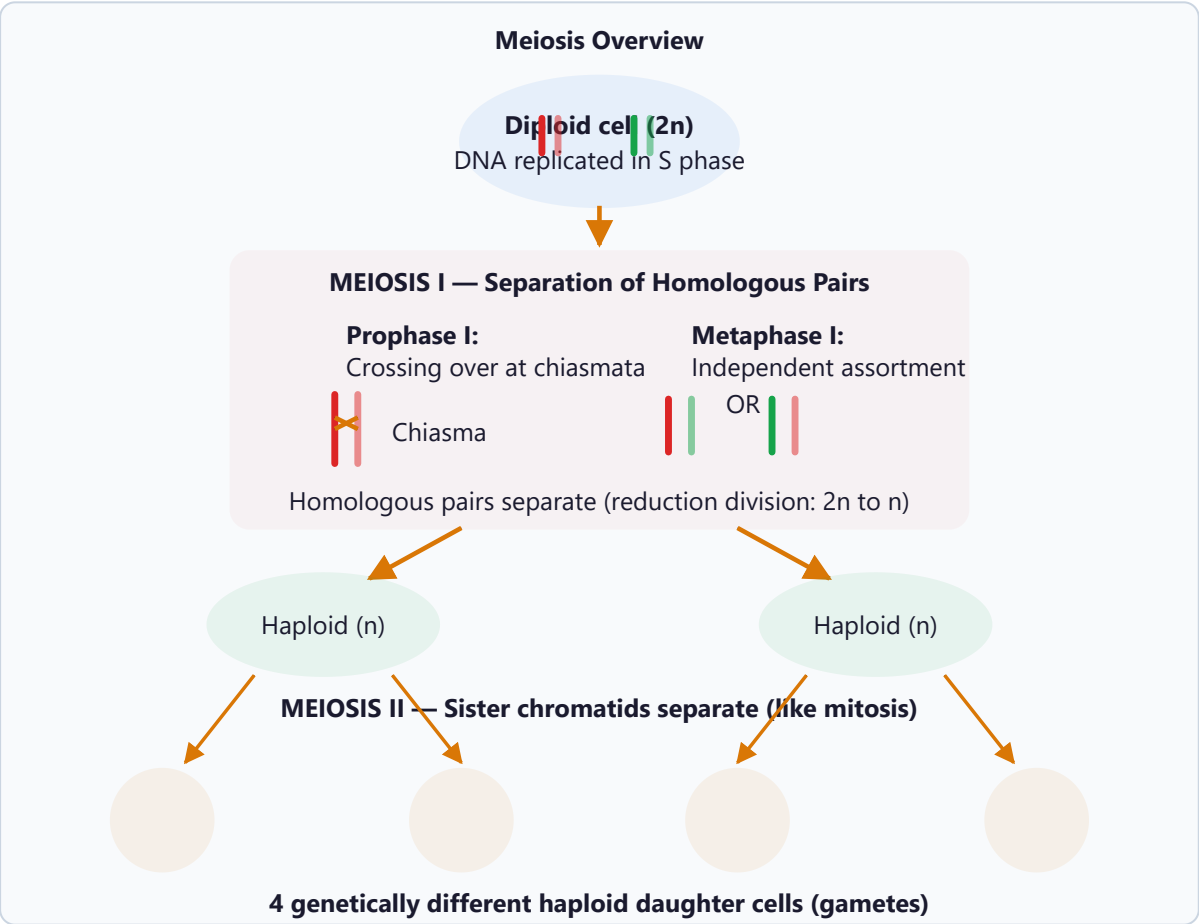
Key Events in Meiosis

Stage	Key Events	Significance
Prophase I	Homologous chromosomes pair up forming **bivalents**. Non-sister chromatids exchange segments at **chiasmata** (crossing over)	Produces recombinant chromosomes with new allele combinations
Metaphase I	Bivalents line up randomly at the cell equator (**independent assortment**)	Random orientation means each gamete gets a different combination of maternal/paternal chromosomes
Anaphase I	Homologous pairs separate (one from each pair to each pole). Centromeres do NOT divide	Reduction division: chromosome number halved (2n to n)
Meiosis II	Sister chromatids separate (centromeres divide). Similar to mitosis but with haploid cells	Produces 4 haploid cells from the 2 cells made in meiosis I

Sources of Genetic Variation in Meiosis

- **Crossing over (prophase I):** Homologous chromosomes form bivalents; non-sister chromatids exchange genetic material at chiasmata, producing **recombinant chromosomes** with new combinations of alleles. This occurs at random positions.
- **Independent assortment (metaphase I):** Homologous pairs line up randomly at the cell equator. The orientation of each pair is independent of all others. With n pairs of chromosomes, there are $\mathbf{2^n}$ possible combinations of chromosomes in the gametes. In humans (n = 23), this gives 2^{23} = 8,388,608 combinations.
- **Random fertilisation:** Any sperm can fuse with any egg, multiplying possible genetic combinations. In humans: 8.4 million x 8.4 million = over 70 trillion combinations (before considering crossing over).

$$\text{Total combinations from independent assortment alone} = 2^{\text{n}} \text{ (where n = haploid number)}$$



Exam Tip: Be clear about the difference between meiosis I and meiosis II. In meiosis I, homologous pairs separate (reduction division) — this is where independent assortment and crossing over introduce variation. In meiosis II, sister chromatids separate — this is essentially a mitotic division of

haploid cells. A common mistake is saying "chromosomes separate" without specifying which type of separation occurs at each stage.

Comparison: Mitosis vs Meiosis

Feature	Mitosis	Meiosis
Number of divisions	1	2
Daughter cells produced	2 (diploid, genetically identical)	4 (haploid, genetically different)
Crossing over	No (rare)	Yes (prophase I)
Independent assortment	No	Yes (metaphase I)
Bivalent formation	No	Yes (prophase I)
Function	Growth, repair, asexual reproduction	Gamete production
Chromosome number	Maintained (2n to 2n)	Halved (2n to n)

4.6 Genetic Diversity and Adaptation

Genetic diversity: The total number of different alleles in a population. It can be measured as the proportion of genes that have more than one allele, or as the heterozygosity index.

Factors that increase genetic diversity:

- **Mutations:** The ultimate source of new alleles. Most are neutral or harmful, but some are beneficial — essential raw material for evolution.
- **Gene flow (migration):** Movement of alleles between populations through migration. Introduces new alleles and increases diversity.
- **Sexual reproduction:** Meiosis (crossing over + independent assortment) and random fertilisation produce new combinations of alleles each generation.

Factors that decrease genetic diversity:

- **Genetic drift:** Random changes in allele frequency due to chance events, especially significant in small populations. Can lead to fixation (allele frequency = 1) or loss of alleles.
- **Genetic bottlenecks:** A dramatic reduction in population size (e.g. disease, natural disaster) randomly eliminates alleles. The surviving population has reduced genetic diversity. Example: cheetahs passed through a bottleneck ~10,000 years ago and now have extremely low genetic diversity — making them vulnerable to disease.
- **Founder effect:** A small group colonises a new area. The founding population carries only a subset of alleles from the original population, so allele frequencies differ. Example: Afrikaner population in South Africa has high frequency of Huntington disease due to founder effect.
- **Selective breeding (artificial selection):** Humans select for desirable traits, reducing allele diversity. Inbreeding depression can result.
- **Directional natural selection:** Favours one phenotype extreme, shifting allele frequencies and reducing diversity.

Adaptation

Adaptation: A feature that increases an organism's fitness (ability to survive and reproduce in its environment). Adaptations arise through natural selection acting on genetic variation.

Type	Definition	Example
Anatomical	Physical/structural feature	Thick fur of Arctic fox for insulation; long roots of desert plants to reach deep water
Physiological	Biochemical or metabolic feature	Antifreeze proteins in Antarctic fish; production of venom by snakes
Behavioural	An action that aids survival	Migration of birds to warmer climates; courtship displays for mate selection

4.7 Taxonomy and Classification

Taxonomy: The science of classifying organisms into hierarchical groups based on shared characteristics. The Linnaean hierarchy from broadest to most specific: **Domain, Kingdom, Phylum, Class, Order, Family, Genus, Species**.

Mnemonic: **D**ear **K**ing **P**hilip **C**ame **O**ver **F**or **G**ood **S**paghetti

Species: A group of organisms that can interbreed to produce fertile offspring and are reproductively isolated from other groups. (Note: this biological species concept has limitations — it cannot apply to asexual organisms or fossils.)

The Three-Domain System (Carl Woese, 1990)

Based on rRNA sequence comparisons, Woese proposed classifying life into three domains:

Domain	Cell Type	Key Features
Bacteria	Prokaryotic	Peptidoglycan cell walls; ester-linked membrane lipids; unique rRNA sequences
Archaea	Prokaryotic	No peptidoglycan (pseudopeptidoglycan or protein S-layers); ether-linked lipids; many are extremophiles (thermophiles, halophiles, methanogens); rRNA sequences more similar to Eukarya than to Bacteria
Eukarya	Eukaryotic	Membrane-bound nucleus and organelles; linear chromosomes; 80S ribosomes. Includes animals, plants, fungi, protists

Methods of classification:

- **Morphology/anatomy:** Traditional approach using observable features. Limited because convergent evolution can produce similar features in unrelated organisms (analogous structures) and environmental variation can alter appearance.
- **Molecular phylogenetics:** Comparing DNA base sequences, mRNA sequences, or amino acid sequences. More objective and reliable than morphology. Fewer differences = more closely related (more recent common ancestor). Used to construct phylogenetic trees.
- **Immunological comparisons:** Antibodies raised against proteins of one species are mixed with proteins of another. The degree of **precipitin reaction** (antibody-antigen binding) indicates how similar the proteins are. More precipitation = more similar proteins = more closely related.

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- **Courtship behaviour:** Species-specific courtship rituals act as reproductive isolating mechanisms and can help identify closely related species that look similar.

Exam Tip: Molecular phylogenetics is considered more reliable than morphology because it is less subjective, not affected by convergent evolution, and provides quantitative data. DNA sequences change at a roughly constant rate (molecular clock), allowing estimation of when species diverged.

4.8 Biodiversity

Biodiversity: The variety of living organisms in an area, considered at three levels:

- **Genetic diversity:** Variation in alleles within a species
- **Species diversity:** The number of different species AND their relative abundance (evenness) in a community
- **Ecosystem diversity:** The range of different habitats, communities and ecological processes in an area

Species Richness vs Species Diversity

Species richness is simply the number of different species present. However, it does not account for relative abundance — a community with 100 species where one species dominates 99% of individuals is less diverse than one with 100 equally abundant species. The **species diversity index** accounts for both richness and evenness.

Simpson's Diversity Index

$$D = 1 - [\Sigma(n/N)^2] \quad \text{or equivalently} \quad d = N(N - 1) / \Sigma n(n - 1)$$

Where N = total number of organisms of all species, n = total number of organisms of each species.

- In the D version: values range from 0 to 1, where 1 = maximum diversity.
- In the d version: higher values indicate greater diversity; there is no upper limit.
- A community dominated by one species will have a low diversity index.

Sampling Methods

Method	Used For	Key Points
Random quadrats	Sessile (non-moving) organisms — plants, barnacles	Use random number generator for coordinates to avoid bias. Count or estimate % cover. Repeat many times for reliability.
Line transect	Changes across a gradient (e.g. seashore zonation)	Record species touching a line at regular intervals
Belt transect	More detailed gradient study	Quadrats placed at intervals along the line
Mark-release-recapture	Mobile animals	Lincoln index: Population = $(n_1 \times n_2) / n_m$ Assumptions: no births/deaths/migration between samples; marks do not affect survival; marks do not rub off; sufficient time for mixing

Exam Tip: When calculating the diversity index, a common error is forgetting that n(n-1) is calculated for EACH species separately, then the values are summed. Practice with worked examples. Also remember that farming monocultures reduce biodiversity — this is a frequent extended response topic. Link to deforestation, habitat fragmentation, and the importance of conservation.

Human Activities Affecting Biodiversity

- **Deforestation:** Removes habitats; reduces species diversity; releases stored carbon (contributing to climate change)
- **Agriculture:** Monocultures reduce plant diversity; pesticides reduce insect diversity; hedgerow removal destroys habitats
- **Pollution:** Eutrophication (excess fertiliser) reduces aquatic species diversity; air pollution (acid rain) damages ecosystems
- **Climate change:** Shifts habitats faster than many species can migrate; ocean acidification threatens coral reefs

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- **Overexploitation:** Overfishing, hunting, and poaching directly reduce population sizes

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